# What the shelf check costs

Sensei challenge, JunctionX Lisbon 2026. Cost of the pipeline as frozen for the validation run (Claude Opus 5.5, one answer per planogram area), what it cost to build, and what it would cost in a store. All figures come from the tokens recorded on every saved answer, at Claude API list prices. 11 October 2026.

**$3.20**
the validation run: 26 photos, 289 areas, about one minute

**$0.12**
per photo, or $0.011 per planogram area

**$21.70**
one store day: 14 cameras, 181 photos, 1,962 areas

**$157**
spent on the API to build and test it, all models

## One answer per area

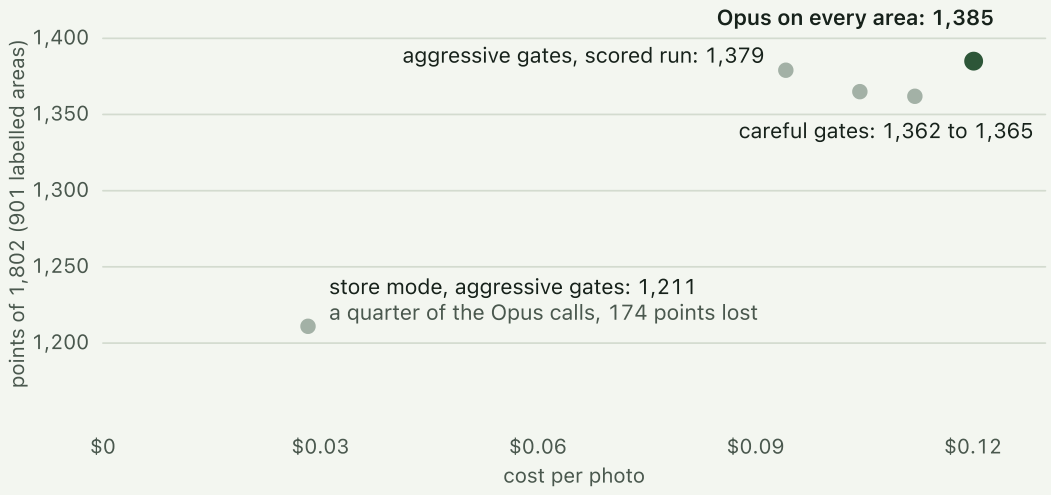
Each planogram area is one Opus call: the whole photo with the area outlined, a close-up and the area straightened, plus the prompt. The system prompt is cached, so most of the input after the first call is billed at the cache rate.

| Per Opus call, frozen config (average of 1,829 calls) | Tokens | Price per million | Cost |
|---|---|---|---|
| Input (images + prompt, incl. cache reads and writes) | 3,919 | $0.20 to $5.00 | $0.0087 |
| Output (the JSON verdict) | 120 | $20.00 | $0.0024 |
| **One area** | | | **$0.0111** |

Opus 5.5 list prices per million tokens: input $4.00, output $20.00, cache read $0.20, cache write $5.00. A camera sees 6 to 16 areas (11 on average), hence $0.12 per photo.

## Coded rules before Opus: cheaper, never better

The gates (covered shelf, nothing changed, clearly empty, clearly full with a colour fingerprint of the product) answer an area themselves when they are sure, and only the rest goes to Opus. Replayed offline on the 181 photos of the training day with saved Opus answers: no API call was made for this test.




For the validation run the cost question is settled: $3.20 buys the best score we can measure, and every cheaper route we tried gives up points.

## The gate replay in numbers

| Setting | All 901 areas | Clean 406 | Opus calls | Per photo | Store day |
|---|---|---|---|---|---|
| **Opus on every area (frozen config)** | **1,385** | **576** | **100 %** | **$0.120** | **$21.74** |
| Scored run, aggressive gates | 1,379 | 562 | 78 % | $0.094 | $17.04 |
| Store mode, careful gates | 1,365 | 557 | 87 % | $0.104 | $18.89 |
| Scored run, careful gates | 1,362 | 551 | 93 % | $0.112 | $20.27 |
| Store mode, aggressive gates | 1,211 | 489 | 24 % | $0.028 | $5.12 |

"Clean 406" are the areas on 64 photos that never fed a threshold, the fairest check. The clearly-full gate wins on the labels it was tuned on and loses 14 to 25 points on the clean ones; no clearly-empty setting qualifies. The colour fingerprint tells right from wrong product well where it can be measured, but every Opus mistake on product sits on a near-empty area where it cannot, so it adds no points. Store mode reuses an earlier photo of the same day, which the scored run may not do.

## Model choice

The same 10 evening photos (109 areas, at most 218 points) through each model. This study ran with reference pictures in the prompt, which made every call dearer than today; the last row is the frozen config on the same photos.

| Configuration | Points of 218 | False product alarms | Per photo |
|---|---|---|---|
| Haiku, one answer | 95 | 10 | $0.004 |
| Haiku, three answers | 97 | 6 | $0.012 |
| Fine-tuned Qwen 4B on the laptop | 102 | n/a | $0 API |
| Sonnet, one answer | 126 | 2 | $0.08 |
| Haiku first, Opus where unsure | 127 | 0 | $0.04 |
| Opus, three answers | 140 | 0 | $0.45 |
| Sonnet first, Opus where unsure | 143 | 0 | $0.29 |
| **Opus, one answer, photo only (frozen)** | **145** | **0** | **$0.12** |

The cheap models read stock levels reasonably but raise false wrong-product alarms, each worth −2. The local model costs nothing per call but needs 20 to 40 minutes for a run on this laptop and prints the same confidence on almost every area, so a cascade cannot tell when to hand over to Opus.

## What the build cost

| Model | Answers | Cost |
|---|---|---|
| Opus 5.5 | 12,613 | $150.90 |
| Sonnet 5.5 | 708 | $4.76 |
| Haiku 5.5 | 4,773 | $1.49 |
| Local models | 1,073 | $0 |
| **Total** | **19,167** | **$157.15** |

From the saved answers. Calls that failed or timed out are not saved, so this is a floor; the real bill is a little higher. Haiku includes 1,697 untagged answers from the Haiku-only prototype.

## In a store

| Per store, 30 days | Cost |
|---|---|
| **Opus on every area, hourly** | **$652** |
| Careful gates | $567 |
| Aggressive gates | $154 |

At the training day's rhythm: 14 cameras over fruit and vegetables, a photo roughly every hour. Gates pay off when photos come every few minutes, because then most areas really have not changed since the last one.

Method: tools/gate_replay_now.py, out/experiments.jsonl, docs/decisions.md